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$$\begin{aligned}& \int \frac{x}{\cos^2 x} dx \\& \left\{ \begin{array}{l} u = x \\ dv = \frac{1}{\cos^2 x} dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = dx \\ v = \tan x \end{array} \right. \\& \int \frac{x}{\cos^2 x} dx \\& = x \tan x - \int \tan x dx \\& = x \tan x - \int \frac{\sin x}{\cos x} dx \\& u = \cos x \Rightarrow du = -\sin x dx \\& = x \tan x + \int \frac{1}{u} dx \\& = x \tan x + \ln |\cos x| + C\end{aligned}$$

References